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Background flow method for Vlasov Equation

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Abstract

This study applies the background flow method, a special case of the auxiliary functional technique, to the Vlasov–Maxwell system with the aim of deriving rigorous bounds on discharge current in Hall thrusters. Starting from steady-state solutions under constant electromagnetic fields, Lyapunov functionals are used to establish linear and nonlinear stability. The analysis is then extended to spatially varying fields through potential-based models, yielding explicit expressions for steady distributions, fields, and current density. These results lay the foundation for applying the background flow framework to obtain mathematically rigorous performance bounds for plasma propulsion systems.

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1 Introduction

In kinetic theory, a long-standing difficulty lies in establishing reliable bounds for physically relevant quantities in systems described by the Vlasov equation. Numerical simulations of collisionless plasmas can, in principle, resolve the intricate dynamics of such systems, yet the computational demands become overwhelming in high-dimensional phase spaces or when extremely long time integrations are required. An alternative route is to bypass the full solution and instead develop analytical estimates that place rigorous limits on the long-time averages of selected observables. These bounds offer firm theoretical constraints even when numerical methods may suffer from resolution limits or accumulated errors. Among the techniques available, the auxiliary functional method—particularly in its specialized form known as the background flow method—has shown notable success in constraining quantities in both fluid and plasma models without requiring complete knowledge of the evolving state. The background flow method can, in fact, be interpreted as a specific case of the auxiliary functional framework in which the auxiliary functional is quadratic in form [1].

This report builds on the work of [2], who applied the linear auxiliary functional method to a model problem derived from the Vlasov–Poisson equation, successfully obtaining rigorous upper and lower bounds on the current. The aim of the present study is to follow a similar framework, but extend its application to the full Vlasov–Maxwell system. The Vlasov–Maxwell system can be regarded as a generalization of the Vlasov–Poisson model obtained by relaxing the electrostatic approximation and retaining the full set of Maxwell equations. In this work, we consider the case of a single-particle species in a three-dimensional configuration, subject to idealized boundary conditions.

In terms of physical applications, we focus on the plasma discharge in a Hall effect thruster (HET). A Hall thruster is an electric propulsion device that accelerates ions using a combination of a radial magnetic field and an axial electric field, while electrons are magnetically confined in the azimuthal direction. The device consists of an annular discharge channel, typically made of an insulating material, into which a neutral propellant gas (such as xenon) is injected through the anode. Surrounding magnetic coils generate a predominantly radial magnetic field near the channel exit, and an external cathode positioned outside the channel emits electrons into the discharge region. Inside the channel, the neutral atoms injected are ionized through collisions with these energetic electrons. The applied axial electric field then accelerates the resulting positively charged ions toward the exit of the channel, generating thrust, while the radial magnetic field strongly impedes the motion of the axial electrons, but allows the azimuthal drift, giving rise to the Hall current [3]. A schematic diagram of this configuration is shown in Fig. 1.

A key performance metric in Hall thruster operation is the steady-state discharge current, I, as it has a direct impact on the thrust, T, generated by the device. In its simplest form, the thrust can be expressed as

$$T = \dot{m}\, v_{\mathrm{ex}}, \tag{1.1}$$

where $\dot{m}$ is the propellant mass flow rate and v_{ex} is the ion exhaust velocity. Assuming that ions are accelerated through a potential drop V_a and neglecting losses [5], energy conservation gives

$$\frac{1}{2} m_i v_{\mathrm{ex}}^2 = q V_a, \tag{1.2}$$

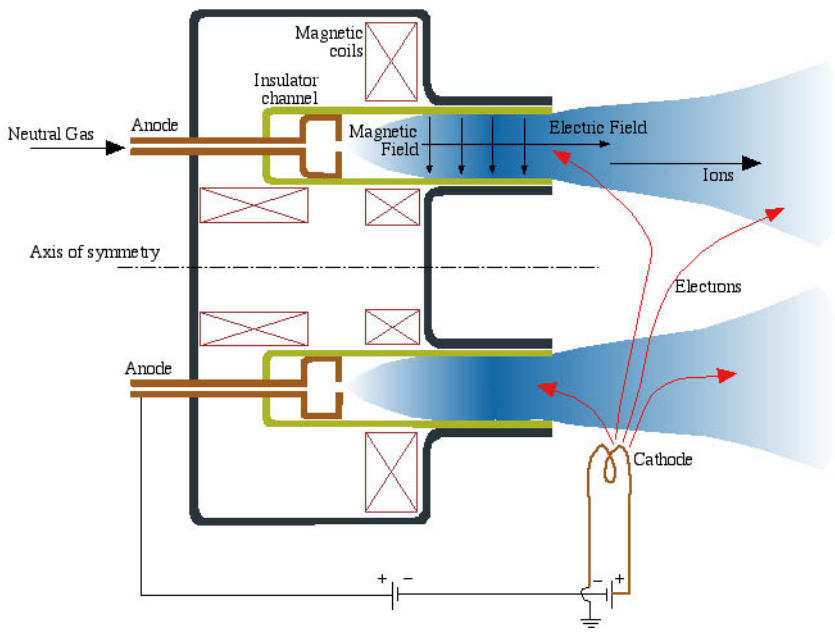


Figure 1: This is a schematic diagram of a Hall-effect thruster. Neutral gas is injected through the anode into the discharge channel, where it is ionized by electrons emitted from the cathode. The magnetic coils produce a radial magnetic field, and the applied electric field accelerates the ions axially to produce thrust. Electrons undergo azimuthal drift due to the crossed $\mathbf{E}$ and $\mathbf{B}$ fields, forming the Hall current [4].

where m_i and q are the ion mass and charge, respectively. Combining Eqs. (1.1) and (1.2), and noting that $I = q\,\dot{N}$ with $\dot{N}$ the ion production rate, yields [5]

$$T \approx I\sqrt{\frac{2m_i}{q}V_a}. \tag{1.3}$$

Equation (1.3) shows that, under idealized assumptions, the thrust scales linearly with the discharge current for a given acceleration voltage. Therefore, obtaining rigorous bounds on I directly translates into predictive bounds on the achievable thrust, providing valuable performance constraints for Hall thruster design and optimization.

2 Background

2.1 The Auxiliary Functional Method

The auxiliary functional method is a variational technique used to derive rigorous bounds on infinite-time averages of physical quantities governed by partial differential equations (PDEs), without requiring the equations to be solved explicitly [1]. Originally developed in the context of turbulent fluid flows [6], it has since been adapted to plasma physics and kinetic theory, particularly for systems such as the Vlasov–Poisson equations. Let $\xi(t)$ represent the evolving state of the system in an appropriate function space, for example, the distribution function $f(x, v, t)$, and possibly the electromagnetic fields $E(x, t)$, $B(x, t)$ in the Vlasov–Maxwell system. The dynamics of $\xi(t)$ is governed by a nonlinear PDE of the form

$$\frac{\mathrm{d}\xi}{\mathrm{d}t} = \mathcal{N}[\xi]\,, \tag{2.1}$$

where $\mathcal{N}$ is a nonlinear differential operator. We define a physical quantity of interest $\Phi[\xi]$, whose infinite-time average we aim to bound:

$$\overline{\Phi} = \limsup_{T\to\infty} \frac{1}{T}\int_0^T \Phi[\xi(t)]\ \mathrm{d}t. \tag{2.2}$$

The central idea of the auxiliary functional method is to construct an auxiliary functional $V[\xi]$, differentiable along solutions $\xi(t)$, such that its time derivative (Lie derivative) is

$$\frac{\mathrm{d}V}{\mathrm{d}t} = \left\langle \frac{\delta V}{\delta \xi}, \frac{\mathrm{d}\xi}{\mathrm{d}t} \right\rangle = \left\langle \frac{\delta V}{\delta \xi}, \mathcal{N}[\xi] \right\rangle, \tag{2.3}$$

where $\langle \cdot, \cdot \rangle$ denotes the appropriate inner product. If $V[\xi(t)]$ remains bounded in time, the time average of its derivative satisfies

$$\lim_{T\to\infty} \frac{1}{T}\int_0^T \frac{\mathrm{d}V}{\mathrm{d}t}\,\mathrm{d}t = 0, \tag{2.4}$$

so that we obtain the key balance identity,

$$\overline{\left\langle \frac{\delta V}{\delta \xi}, \mathcal{N}[\xi] \right\rangle} = 0. \tag{2.5}$$

The auxiliary functional is chosen so that the inner product in (2.5) explicitly involves the quantity of interest. In particular, one arranges

$$\left\langle \frac{\delta V}{\delta \xi}, \mathcal{N}[\xi] \right\rangle = \Phi[\xi] + R[\xi], \tag{2.6}$$

where $R[\xi]$ denotes the remainder terms depending on the choice of V. Then averaging in time gives

$$\overline{\Phi[\xi]} + \overline{R[\xi]} = 0. \tag{2.7}$$

At this stage, functional inequalities such as Cauchy–Schwarz or Hölder [1] are applied to bound the remainder:

$$R[\xi] \ \geq\ -\mathcal{F}[V, \xi], \tag{2.8}$$

where $\mathcal{F}$ depends only on the auxiliary functional and known system quantities. Substituting into (2.7) yields the desired estimate

$$\overline{\Phi} \;\leq\; \mathcal{F}[V, \xi]. \tag{2.9}$$

The final step of the method is to optimize the admissible class of auxiliary functionals V, thus obtaining the sharpest possible bound on the long-term average $\overline{\Phi}$.

2.2 Background Flow Method.

The background flow method is a special case of the auxiliary functional method, used to derive rigorous bounds on time-averaged quantities in fluid and plasma systems [1]. It is based on introducing a carefully chosen steady vector field, called the background flow, and constructing a fluctuation energy functional whose time derivative along solutions of the governing equations constrains physical observables.

Concretely, the solution $u(x,t)$ is decomposed into a steady background component $b(x)$ and a fluctuation $v(x,t)$:

$$u(x,t) = b(x) + v(x,t). \tag{2.10}$$

The auxiliary functional is then chosen as the fluctuation energy

$$V[u] = \tfrac{1}{2}\|v\|^2 = \tfrac{1}{2}\|u - b\|^2, \tag{2.11}$$

which is equivalent to the shifted form,

$$V[u] = \tfrac{1}{2}\|u\|^2 + \langle b, u\rangle + \text{constant}, \tag{2.12}$$

with $b(x)$ divergence-free and satisfying the boundary conditions of the problem [7].

Differentiating along the trajectories of the PDE

$$\frac{\partial u}{\partial t} = \mathcal{N}[u], \tag{2.13}$$

yields

$$\frac{\mathrm{d}V}{\mathrm{d}t} = \langle v, \mathcal{N}[b+v]\rangle. \tag{2.14}$$

This expression can be decomposed schematically as

$$\frac{\mathrm{d}V}{\mathrm{d}t} = Q[v] + \Phi[u] + R[b], \tag{2.15}$$

where (i) $Q[v]$ is a quadratic form in the fluctuation, (ii) $\Phi[u]$ is the physical quantity of interest, and (iii) $R[b]$ depends only on the background flow.

The method requires the spectral condition

$$Q[v] \;\geq\; 0 \quad \text{for all admissible fluctuations } v, \tag{2.16}$$

which ensures that the contribution of fluctuation cannot destabilize the bound [8]. Since V remains

bounded in time, the averaged derivative vanishes, and one obtains

$$\overline{\Phi[u]} + R[b] \leq 0, \tag{2.17}$$

so that

$$\overline{\Phi[u]} \;\leq\; -R[b]. \tag{2.18}$$

Optimizing over admissible choices of background flow $b(x)$ produces the sharpest possible bounds on the infinite-time average of the quantity of interest. This approach has been successfully applied in fluid dynamics and plasma physics to obtain rigorous estimates without solving the full non-linear PDEs [9].

The method avoids full solution of the governing equations and enables bounding of physically meaningful quantities in complex systems [9].

2.3 Vlasov–Maxwell system

The Vlasov equation governs the evolution of the distribution function $f_s(t, \mathbf{r}, \mathbf{v})$ for each species s in a collisionless plasma. It is given by

$$\frac{\partial f_s}{\partial t} + \mathbf{v} \cdot \nabla_{\mathbf{r}} f_s + \frac{q_s}{m_s} \left[\mathbf{E}(\mathbf{r}, t) + \mathbf{v} \times \mathbf{B}(\mathbf{r}, t)\right] \cdot \nabla_{\mathbf{v}} f_s = S_s(\mathbf{r}, \mathbf{v}, t)\,, \tag{2.19}$$

where q_s and m_s are the charge and mass of species s, respectively [10]. The first term describes the explicit time dependence of f_s, the second term represents advection in physical space, and the third term accounts for acceleration in velocity space due to the Lorentz force. The term $S_s(\mathbf{r}, \mathbf{v}, t)$ is a source term that models processes such as particle injection or ionization. In the present work, the source term will be retained and taken to depend explicitly on velocity $\mathbf{v}$, reflecting the velocity-dependent ionization processes relevant in Hall thrusters.

The electromagnetic fields $\mathbf{E}$ and $\mathbf{B}$ evolve self-consistently according to Maxwell's equations,

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}, \tag{2.20a}$$

$$\nabla \cdot \mathbf{B} = 0, \tag{2.20b}$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \tag{2.20c}$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}, \tag{2.20d}$$

where ε_0 and μ_0 are the vacuum permittivity and permeability, respectively [11]. In electromagnetism, the electric and magnetic fields can be conveniently expressed in terms of potentials. The electrostatic potential ϕ and the magnetic vector potential $\boldsymbol{\psi}$ are defined such that

$$\mathbf{E} = -\nabla\phi - \frac{\partial \boldsymbol{A}}{\partial t}, \qquad \mathbf{B} = \nabla \times \boldsymbol{A}. \tag{2.21}$$

Here $\phi = \phi(\mathbf{x}, t)$ is a scalar function and $\boldsymbol{A} = \boldsymbol{A}(\mathbf{x}, t)$ is a vector field [12]. This representation automatically satisfies $\nabla \cdot \mathbf{B} = 0$ and provides a convenient framework for both steady and time-dependent configurations.

The charge density ρ and current density $\mathbf{J}$ are obtained as velocity moments of the distribution function:

$$\rho(t,\mathbf{r}) = \sum_s q_s \int_{\mathbb{R}^3} f_s(t,\mathbf{r},\mathbf{v})\ \mathrm{d}\mathbf{v}, \quad \mathbf{J}(t,\mathbf{r}) = \sum_s q_s \int_{\mathbb{R}^3} \mathbf{v}\, f_s(t,\mathbf{r},\mathbf{v})\ \mathrm{d}\mathbf{v}. \tag{2.22}$$

In the specific application to Hall thrusters considered here, we focus on a single charged species (the ions) to simplify the problem [13]. The distribution function f depends on both position $\mathbf{r}$ and velocity $\mathbf{v}$, while the fields $\mathbf{E}$ and $\mathbf{B}$ depend only on position and time. This separation reflects the fact that the electromagnetic fields are determined by macroscopic quantities, whereas the distribution function retains full phase-space dependence.

2.4 Lyapunov Functionals and Stability

The concept of a Lyapunov functional provides a systematic way to analyse the stability of dynamical systems such as the Vlasov–Maxwell equations. Given a steady solution ξ_0, one introduces a perturbation $\xi = \xi_0 + g$ and defines a functional

$$L[g] \ \geq\ 0, \qquad L[g] = 0 \ \text{ iff } g = 0, \tag{2.23}$$

that measures the size of the departure from equilibrium [14]. The system is said to be Lyapunov stable if

$$\frac{\mathrm{d}}{\mathrm{d}t} L[g(t)] \ \leq\ 0, \tag{2.24}$$

so that perturbations cannot grow in time.

2.4.1 Linearised Lyapunov stability

Consider a general system

$$\frac{\partial \xi}{\partial t} \ =\ \mathcal{N}[\xi], \qquad \xi(t) \in \mathcal{X}, \tag{2.25}$$

with steady state ξ_0 satisfying $\mathcal{N}[\xi_0] = 0$. Expanding around ξ_0 with $\xi = \xi_0 + \varepsilon\eta$ gives

$$\frac{\partial \eta}{\partial t} \ =\ \mathcal{L}\eta \ +\ \mathcal{Q}(\eta,\eta), \tag{2.26}$$

where $\mathcal{L}$ is the linearisation (Fréchet derivative) of $\mathcal{N}$ at ξ_0, and $\mathcal{Q}(\eta,\eta)$ contains second–order and higher perturbation terms [15]. Linearised Lyapunov analysis corresponds to neglecting $\mathcal{Q}$, leaving

$$\frac{\partial \eta}{\partial t} = \mathcal{L}\eta. \tag{2.27}$$

In this setting one seeks a quadratic functional $L[\eta]$ such that

$$\frac{\mathrm{d}}{\mathrm{d}t} L[\eta(t)] = \left\langle \frac{\delta L}{\delta \eta},\, \mathcal{L}\eta \right\rangle \ \leq\ 0. \tag{2.28}$$

If $\dot{L} \leq -\mu\|\eta\|^2$ for some $\mu > 0$, perturbations decay exponentially [16]. Thus linear Lyapunov stability ensures that the system locally returns to the steady state when perturbations are small. Nonlinear Lyapunov stability requires controlling the dropped $\mathcal{Q}$ terms as well.

2.4.2 Connection to the background–flow framework

The Lyapunov approach establishes the starting point: find a quadratic functional $L[g]$ such that its time derivative along solutions is non–positive,

$$\frac{d}{dt}L[g(t)] \;\leq\; 0. \tag{2.29}$$

This guarantees stability, but does not yet produce quantitative information such as bounds on fluxes or currents.

The background–flow (or auxiliary functional) method extends this idea. The key step is to add cross terms involving a chosen steady background profile U_{bg}, leading to a modified functional

$$L[g] \;=\; \int |g|^2\,dx \;+\; \int g\,U_{\text{bg}}\,dx. \tag{2.30}$$

Differentiating (2.30) in time gives

$$\frac{d}{dt}L[g] \;=\; \underbrace{\frac{d}{dt}\int |g|^2 dx}_{\text{basic Lyapunov part}} \;+\; \underbrace{\frac{d}{dt}\int gU_{\text{bg}}dx}_{\text{cross term}}. \tag{2.31}$$

The role of the cross term is crucial: it can be chosen so that all conservative contributions cancel exactly [17]. This leaves

$$\frac{d}{dt}L[g] \;\leq\; \mathcal{D}[g], \tag{2.32}$$

where $\mathcal{D}[g]$ depends only on dissipative or source terms, i.e. the controllable parts of the dynamics. From this inequality one can then extract an estimate of the form

$$\limsup_{T\to\infty}\frac{1}{T}\int_0^T F[g(t)]\,dt \;\leq\; C, \tag{2.33}$$

where $F[g]$ is a physical quantity of interest (such as current or energy flux) and C is an explicit constant.

In this way, the auxiliary functional method generalises the Lyapunov idea:

- Lyapunov analysis ensures stability by requiring $\dot{L} \leq 0$.
- Cross terms are then introduced to eliminate neutral conservative parts.
- The resulting inequality leads directly to quantitative bounds of the form (2.33).

Thus, Lyapunov stability provides the guiding principle, while the background–flow framework builds on it to obtain rigorous estimates for macroscopic quantities. In the plasma setting, this will ultimately be applied to bounding the steady electric current.

3 Aim and Objectives

3.1 Aim.

The ultimate goal of this project is to establish rigorous mathematical bounds on the discharge current in Hall thrusters, described within the Vlasov–Maxwell framework. Formally, the target is to obtain inequalities of the type

$$\mathcal{L}\big[f_0, E_0, B_0, S\big] \;\leq\; I \;\leq\; \mathcal{U}\big[f_0, E_0, B_0, S\big], \tag{3.1}$$

where I denotes the total discharge current, and the bounding functionals $\mathcal{L}$ and $\mathcal{U}$ depend only on prescribed inputs of the model such as the distribution function, the electromagnetic fields, and the source structure. Such bounds provide a rigorous envelope for the possible current values admitted by the kinetic system, and hence give direct constraints on thruster performance without requiring full numerical solution of the governing equations.

3.2 Objectives

To advance toward the overarching aim of deriving rigorous bounds on discharge current in Hall thrusters, this project will be organized around the following **objectives**:

- **Construct and analyse candidate steady states.** Try to formulate explicit steady-state backgrounds (f_0, E_0, B_0, S_0) of the Vlasov–Maxwell system under controlled simplifying assumptions. Two background configurations will be used as scaffolding:

  1. *Constant-field baseline:* Specify a uniform magnetic field and the simplest consistent electric field and source, together with a drift-free reference distribution. This baseline will serve as a clean operating point for subsequent stability analysis.
  2. *x-dependent model:* Parameterise spatially varying fields through prescribed potentials and a drift profile, together with an assumed source structure (constant and velocity-dependent channels). This model will clarify how potentials and drift map to fields and current in more realistic settings.

- **Establish a stability framework.** Our primary goal is to prove nonlinear(Lyapunov) stability of the chosen steady state by proposing an energy-like functional and showing its time derivative is nonpositive under the intended boundary and source conditions. If sign–indefinite couplings make this intractable, we will pass to the linearised system to avoid the complicated terms, using the linear result to diagnose which cross terms (e.g., background–flow augmentation) are needed. When a full Lyapunov proof succeeds, the steady state is stable for the complete system, and trajectories starting near it converge back to that state: the steady profile is the eventual long-time limit. If only linear stability can be certified, we will state its regime of validity and use it to guide the construction of auxiliary functionals toward quantitative bounds.

- **Translate stability into quantitative bounds.** Build auxiliary functionals anchored to the chosen steady states in order to convert the energy balance into inequalities that bound the discharge current. Make explicit how the bounds depend on modeling choices (potentials, drift), source assumptions, and boundary conditions relevant to Hall-thruster channels.

- **Scope and verification.** Benchmark the workflow on canonical domains and parameter choices, check consistency with conservation laws, and outline how the methodology extends to device-relevant geometries and operating regimes.

Together, these objectives define a unified workflow: select a consistent steady background, analyze its stability, augment the functional to neutralize non-conservative couplings, and use the resulting structure to derive rigorous, interpretable bounds on current in Hall-thruster-like configurations.

4 Boundary conditions for the Hall thruster set-up

In order to ensure a mathematically well-posed Vlasov–Maxwell initial–boundary value problem with finite macroscopic moments (density, current, energy) and physically consistent fluxes, appropriate boundary conditions are imposed both on the steady state $(f_0, E_0, B_0, \phi, \boldsymbol{A})$ and on the perturbations (f', E', B'). These are summarised below.
Let $\mathbf{n}$ be the outward unit normal on $\partial\Omega_x$, $v_n = \mathbf{v}\cdot\mathbf{n}$, and $f(t, \mathbf{x}, \mathbf{v})$ the distribution function. The electromagnetic fields are $(\mathbf{E}, \mathbf{B})$, with electrostatic and magnetic potential ϕ ,$\boldsymbol{A}$ when relevant. The spatial domain is taken as

$$\Omega_x = [0, L] \times [0, r] \times \mathbb{R}, \tag{4.1}$$

where $x \in [0, L]$ is the axial coordinate, $y \in [0, r]$ the radial coordinate, and $z \in \mathbb{R}$ the homogeneous direction in which all quantities are constant.

4.1 Boundary Conditions on steady solution and $(f_0, E_0, B_0, \phi, \boldsymbol{A})$

Kinetic boundary conditions.

- **Anode (metal, upstream).** Charged particles are absorbed,which can be described in the formula below

$$f_0(t, 0, \mathbf{v}) = 0, \qquad v_n < 0. \tag{4.2}$$

  . The electric potential is fixed to $\phi = \phi_a$ (Dirichlet condition). A Dirichlet boundary condition fixes the variable's value on the boundary. We use it when the hardware enforces a known value—like a set voltage or specified particle source—so the model matches the device and the math problem is well- posed [18].

- **Cathode (external emitter).** Electrons are injected as a half–Maxwellian with prescribed current and temperature; incoming particles are absorbed.

- **Dielectric walls.** Charged particles undergo specular reflection,

$$f(\mathbf{x}, v_{\text{out}}) = f(\mathbf{x}, v_{\text{in}} - 2v_n\mathbf{n}), \quad v_n < 0, \tag{4.3}$$

  with optional secondary electron emission (SEE) of yield $\delta(E, \theta)$. This condition makes sure that all the particles will not escape the domain from the wall which indicate no loss exist except the exit.

- **Exit / plume.** Outflow condition: no inflow of charged particles,,which can be described in the formula below

$$f_0(t, L, \mathbf{v}) = 0, \qquad v_n > 0. \tag{4.4}$$

Behaviour in velocity space The perturbation distribution must decay sufficiently fast for the integrability of all velocity moments:

$$\lim_{|\mathbf{v}|\to\infty} f_0(\mathbf{x}, \mathbf{v}) = 0, \tag{4.5}$$

This condition ensures that the distribution function does not diverge as the velocity tends to infinity, thereby avoiding unphysical infinite particle quantities.

Electrostatic special cases. For slab geometry $x \in [0, L]$ with grounded ends $\phi(0) = \phi(L) = 0$. For simplicity, we assume that the equilibrium electric field has only an x–component and depends only on the axial coordinate x. In particular, all transverse components vanish and the electrostatic potential is a function of x alone:

$$\mathbf{E}_0(x) = \big(E_{0x}(x),\, 0,\, 0\big), \qquad \phi = \phi(x), \qquad E_{0x}(x) = -\frac{d\phi(x)}{dx}. \tag{4.6}$$

4.2 Boundary Conditions on Perturbations (f', E', B')

For perturbation quantities, homogeneous boundary conditions consistent with perfectly conducting walls or open boundaries are imposed:

- **Anode (PEC).** For perturbations of the fields,

$$\mathbf{n} \times \mathbf{E}' = 0, \qquad \mathbf{n} \cdot \mathbf{B}' = 0. \tag{4.7}$$

  For perturbations of the distribution,

$$f'(t, 0, \mathbf{v}) = 0 \quad \text{for } v_n < 0. \tag{4.8}$$

- **Exit / plume.** Outflow condition for perturbations $f'(t, L, \mathbf{v})$: no inflow ($f(t, L, \mathbf{v})' = 0$ for $v_n < 0$).
- **Cathode.** Perturbations consistent with emission boundary: set by linearisation of the half–Maxwellian source; fields satisfy PEC conditions.

Behaviour in velocity space. The perturbation distribution must decay sufficiently fast for the integrability of all velocity moments:

$$\frac{f'(\mathbf{x}, \mathbf{v})}{F(|\mathbf{v}|)} \to 0 \quad \text{as } |\mathbf{v}| \to \infty, \tag{4.9}$$

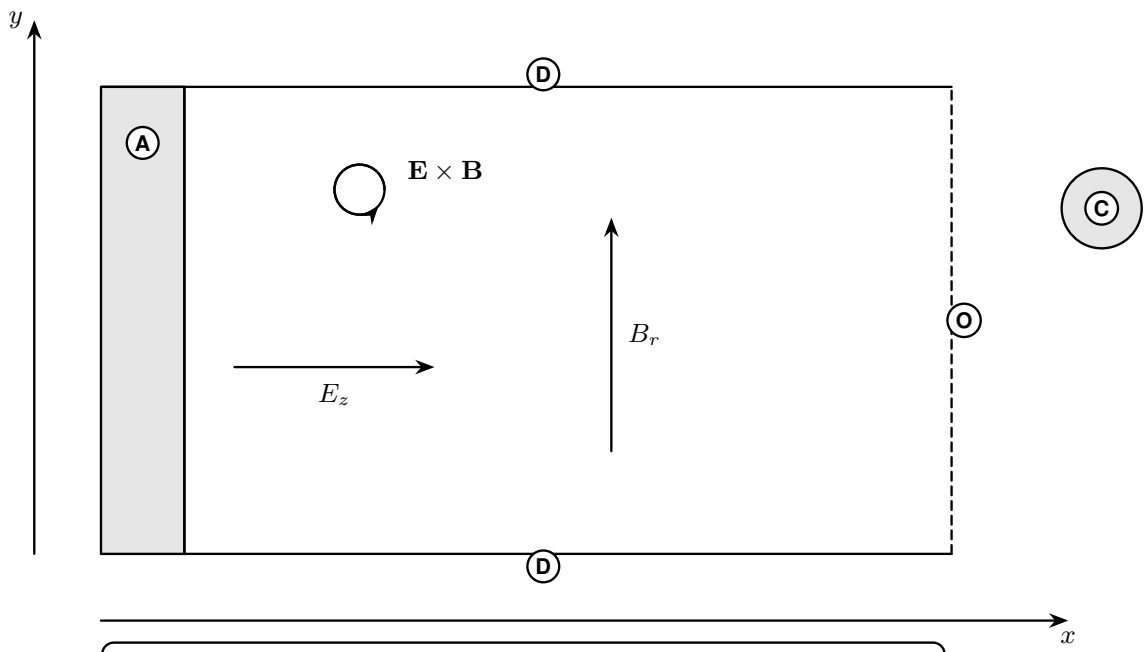
for some Maxwellian-like weight F. In this project we normally take $F(|\mathbf{v}|)$ to be a function of $f_0(t, x, \mathbf{v})$ and this indicates that the perturbation of the distribution does not grow unboundedly with velocity in the analysis.

Condition for Electric and Magnetic field perturbation we assume

$$\partial_t \mathbf{E}' \not\equiv \mathbf{0} \quad \text{and/or} \quad \partial_t \mathbf{B}' \not\equiv \mathbf{0} \qquad \text{on } \Omega_x = [0, L] \times [0, r] \times \mathbb{R}. \tag{4.10}$$

This make sure in the future analysis the time derivative of $\mathbf{E}'$ and $\mathbf{B}'$ will not be zero.

Figure 2 illustrates the Hall-thruster channel and the boundary pieces used in our Vlasov–Maxwell model. The surfaces correspond to: **(A)** anode (metal), **(D)** dielectric channel walls, **(O)** open exit/plume,



Legend (boundary symbols).

A Anode (PEC): $\phi = \phi_a$; $\mathbf{n} \times \mathbf{E}' = 0$, $\mathbf{n} \cdot \mathbf{B}' = 0$; absorbing particles: $f, f' = 0$ for $\mathbf{v} \cdot \mathbf{n} < 0$.

D Dielectric walls (sheath/floating): specular reflection $f(\mathbf{x}, \mathbf{v}_{\text{out}}) = f(\mathbf{x}, \mathbf{v}_{\text{in}} - 2v_n\mathbf{n})$; optional SEE.

O Exit/open plume: outflow for particles: $f, f' = 0$ for $\mathbf{v} \cdot \mathbf{n} > 0$.

C Cathode (PEC): electron injection as half-Maxwellian; fields satisfy PEC ($\mathbf{n} \times \mathbf{E}' = 0$, $\mathbf{n} \cdot \mathbf{B}' = 0$).

Global assumptions: 1D electrostatic steady background $\mathbf{E}_0 = (E_{0x}(x), 0, 0)$, $\phi = \phi(x)$, $E_{0x} = -d\phi/dx$. Velocity-space decay for perturbations: $f'/F(|\mathbf{v}|) \to 0$ as $|\mathbf{v}| \to \infty$ and $\lim_{|\mathbf{v}|\to\infty} f_0(\mathbf{x}, \mathbf{v}) = 0$,

Figure 2: This is the Hall thruster boundary conditions. Symbols A/D/O/C mark the boundary type; details for the boundary condition are given in the legend.

and **(C)** cathode. The arrows indicate the nominal field geometry: axial E_x, radial B_y, and the resulting azimuthal $\mathbf{E} \times \mathbf{B}$ drift.

At each boundary, appropriate conditions are imposed for both particles and fields to ensure a mathematically well-posed initial–boundary value problem, with finite macroscopic moments (density, current, energy) and physically meaningful fluxes. Metals act as absorbing electrodes for particles and as perfect electric conductors (PEC) for fields; dielectric walls enforce specular reflection with optional sheath/secondary-electron-emission (SEE) physics; and the exit is treated as an outflow for particles with an open or radiative boundary for the fields.

5 Steady solution of Vlasov-Maxwell system with constant E-field and B-field

We focus on steady-state solutions of the Vlasov–Maxwell system because many plasma devices, including Hall thrusters, operate in regimes where the macroscopic electromagnetic fields remain approximately constant in time. This can be a simple approximation for the Vlasov Maxwell system in reality. A Lyapunov stable steady state has the property that small perturbations decay over time, so the system naturally returns to its operating point without external control. This makes steady states both physically relevant and mathematically convenient for stability and bound analysis.

The background flow method builds directly on this viewpoint. The full state $(f, \mathbf{E}, \mathbf{B})$ is written as a steady background solution $(f_0, \mathbf{E}_0, \mathbf{B}_0)$, which satisfies the Vlasov–Maxwell equations, plus perturbations $(f', \mathbf{E}', \mathbf{B}')$:

$$f = f_0 + f', \quad \mathbf{E} = \mathbf{E}_0 + \mathbf{E}', \quad \mathbf{B} = \mathbf{B}_0 + \mathbf{B}'. \tag{5.1}$$

Choosing a steady background ensures that all explicit time derivatives of $(f_0, \mathbf{E}_0, \mathbf{B}_0)$ vanish, so the perturbation equations contain only the dynamical effects of $f', \mathbf{E}', \mathbf{B}'$.

Linkage to auxiliary (background–flow) functional. Lyapunov analysis ensures stability by constructing a quadratic functional whose time derivative is nonpositive. However, this only certifies that perturbations cannot grow; it does not yet provide quantitative information such as bounds on fluxes or currents. The auxiliary functional, or background–flow, method extends this idea by adding cross terms involving a steady background profile. These cross terms are chosen to cancel conservative energy exchanges, so that the derivative of the modified functional depends only on dissipative or source contributions. In this way, the method converts the Lyapunov stability condition into inequalities that directly constrain the long-time averages of physical observables like we introduced in 2.4.2.

5.1 Initial conditions

For the purposes of the present analysis, we impose initial conditions in which the background electric and magnetic fields $\mathbf{E}_0$ and $\mathbf{B}_0$ are constant in time, representing the steady operating point of the device. However, we decide to take the electric field to be spatial dependent as if we assign $\mathbf{E}_0$ is spatially uniform, then $\nabla \cdot \mathbf{E}_0 = 0$, and Gauss's law (2.20a) gives

$$\rho_0(\mathbf{r}) = 0. \tag{5.2}$$

Thus, uniform steady fields imply zero charge density everywhere. The charge and current densities are velocity moments of the steady-state distribution function $f_0(\mathbf{r}, \mathbf{v})$:

$$\rho_0(\mathbf{r}) = \sum q \int f_0(\mathbf{r}, \mathbf{v})\, \mathrm{d}\mathbf{v}, \tag{5.3}$$

This requires that the zeroth velocity moment of f_0 yields net neutrality

$$\int f_0(\mathbf{r}, \mathbf{v})\, \mathrm{d}\mathbf{v} = 0 \quad , \tag{5.4}$$

which indicate that f_0 equal to zero under the condition of $\mathbf{E}_0$ is spatially uniform. This will cause meaningless in the problem as this means that there is no particle exist in the domain, therefore we should consider the spatial dependence $\mathbf{E}_0$ to make the model meaningful. This does not imply that the total fields remain fixed, since we allow for non-zero perturbations $\mathbf{E}'$ and $\mathbf{B}'$ that may evolve dynamically. In other words,

$$\mathbf{E}(t, \mathbf{x}) = \mathbf{E}_0 + \mathbf{E}'(t, \mathbf{x}), \quad \mathbf{B}(t, \mathbf{r}) = \mathbf{B}_0 + \mathbf{B}'(t, \mathbf{x}), \tag{5.5}$$

with $\frac{\partial \mathbf{E}_0}{\partial t} = \frac{\partial \mathbf{B}_0}{\partial t} = 0$.

For now, no spatial constraints are imposed on either the background field or perturbations. However, there are some boundary conditions in spatial constraints in the future derivation.

Additionally, the distribution function is decomposed as

$$f(t, \mathbf{x}, \mathbf{v}) = f_0(\mathbf{x}, \mathbf{v}) + f'(t, \mathbf{x}, \mathbf{v}), \tag{5.6}$$

where f_0 is the steady-state background distribution and g is the perturbation.

5.2 Solution for f_0

In SI units, the Ampère–Maxwell and Gauss equations of (2.20d) and (2.20a) can be rewritten into

$$\nabla \times \mathbf{B}_0 = \mu_0 \mathbf{J}_0 + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}_0}{\partial t}, \tag{5.7}$$

$$\nabla \cdot \mathbf{E}_0 = \frac{\rho_0}{\varepsilon_0}, \tag{5.8}$$

where ρ_0 and $\mathbf{J}_0$ are the steady-state charge and current densities.

In steady state, $\partial_t \mathbf{E}_0 = 0$. If $\mathbf{B}_0$ is spatially uniform, $\nabla \times \mathbf{B}_0 = \mathbf{0}$, and (5.7) implies

$$\mathbf{J}_0(\mathbf{r}) = \mathbf{0}. \tag{5.9}$$

Thus, steady fields imply zero current density everywhere.

The current densities are velocity moments of the steady-state distribution function $f_0(\mathbf{r}, \mathbf{v})$:

$$\mathbf{J}_0(\mathbf{r}) = q \int \mathbf{v}\, f_0(\mathbf{r}, \mathbf{v})\, \mathrm{d}\mathbf{v}. \tag{5.10}$$

Equations (5.9)–(5.10) require that the first velocity moment vanishes (no net drift):

$$\int \mathbf{v}\, f_0(\mathbf{r}, \mathbf{v})\, \mathrm{d}\mathbf{v} = \mathbf{0} \quad \text{(no net drift)}. \tag{5.11}$$

A sufficient condition to enforce (5.11) is to choose f_0 to be even in velocity:

$$f_0(\mathbf{r}, \mathbf{v}) = f_0(\mathbf{r}, -\mathbf{v}), \tag{5.12}$$

since the integrand in (5.11) is then odd in $\mathbf{v}$, ensuring its integral vanishes. This symmetry property is often used to construct analytically tractable steady-state backgrounds for stability and bounding analyses.

Based on the above constraints, a natural choice for the steady-state distribution is a spatially uniform Maxwellian of the form

$$f_0(\mathbf{v}) = C\,\mathrm{e}^{-m\mathbf{v}^2}, \tag{5.13}$$

where C is a normalization constant, m is the particle mass, and $v = \|\mathbf{v}\|$.We would assume the distribution function is independent of $\mathbf{x}$ for now. This distribution is isotropic and even in $\mathbf{v}$, ensuring that the first moment condition (5.11) is satisfied identically.

5.3 Solution for the Equilibrium Electric Field E_0

We begin with a simple equilibrium distribution of the form

$$f_0(\mathbf{v}) \;=\; C\,e^{-m|\mathbf{v}|^2}, \qquad m > 0, \tag{5.14}$$

where C is a normalisation constant. This Maxwellian-type ansatz depends only on velocity and is spatially uniform, so the associated number density is constant.

Number density. The number density is given by

$$n_0 \;=\; \int_{\mathbb{R}^3} f_0(\mathbf{v})\,d^3v \;=\; C\int_{\mathbb{R}^3} e^{-m|\mathbf{v}|^2}\,d^3v. \tag{5.15}$$

Using spherical coordinates and the standard Gaussian integral identity

$$\int_{\mathbb{R}^3} e^{-m|\mathbf{v}|^2}\,d^3v = \left(\int_{-\infty}^{\infty} e^{-mv^2}dv\right)^3 = \left(\sqrt{\frac{\pi}{m}}\right)^3 = \left(\frac{\pi}{m}\right)^{3/2}, \tag{5.16}$$

we obtain

$$n_0 = C\left(\frac{\pi}{m}\right)^{3/2}. \tag{5.17}$$

Charge density. The corresponding charge density is constant,

$$\rho_0 \;=\; q\,n_0 \;=\; qC\left(\frac{\pi}{m}\right)^{3/2}. \tag{5.18}$$

Electric field from Gauss's law. Suppose the electric field depends only on x, i.e. $\mathbf{E}(x) = E_x(x)\,\hat{\mathbf{x}}$. Then Gauss's law reduces to

$$\frac{dE_x}{dx} = \frac{\rho_0}{\varepsilon_0}. \tag{5.19}$$

Integrating once yields

$$E_0(x) = \frac{q\,C\left(\frac{\pi}{m}\right)^{3/2}}{\varepsilon_0}\,x + K, \tag{5.20}$$

where K is an integration constant determined by boundary conditions.

Electrostatic potential. By definition $\mathbf{E} = -\nabla\phi$, so

$$E_x(x) = -\frac{d\phi}{dx}. \tag{5.21}$$

Integrating (5.20) gives the potential

$$\phi(x) = \phi(0) - Kx - \frac{\rho_0}{2\varepsilon_0}x^2. \tag{5.22}$$

Here $\phi(0)$ sets the reference level of potential, while the quadratic dependence reflects the uniform charge density. Thus, the equilibrium electric field E_0 follows linearly from Gauss's law with constant charge density, and the associated electrostatic potential $\phi(x)$ is quadratic in x.

From the obtained field to Maxwell–Faraday. We have determined the equilibrium electric field to be

$$\mathbf{E}_0(x) = \big(E_0(x),\,0,\,0\big), \qquad E_0(x) = \frac{\rho_0}{\varepsilon_0}\,x + K. \tag{5.23}$$

Compute its curl in Cartesian coordinates:

$$\nabla \times \mathbf{E}_0 = \begin{pmatrix} \partial_y E_{0,z} - \partial_z E_{0,y} \\ \partial_z E_{0,x} - \partial_x E_{0,z} \\ \partial_x E_{0,y} - \partial_y E_{0,x} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \tag{5.24}$$

since $E_{0,y} = E_{0,z} = 0$ and $E_{0,x} = E_0(x)$ depends only on x. By Maxwell–Faraday,

$$\nabla \times \mathbf{E}_0 = -\,\partial_t \mathbf{B}_0, \tag{5.25}$$

so (5.24) implies $\partial_t \mathbf{B}_0 = \mathbf{0}$: the magnetic field is constant in time. This verifies our steady-state assumption on $\mathbf{B}_0$. (Spatial variations of $\mathbf{B}_0$ are governed solely by $\nabla \cdot \mathbf{B}_0 = 0$ and the boundary conditions of the model.)

5.4 Steady source S_0 consistent with a prescribed f_0

Starting from the Vlasov equation with constant fields,

$$\frac{\partial f}{\partial t} + \mathbf{v}\cdot\nabla_{\mathbf{x}} f + \frac{q}{m}\Big(\mathbf{E}_0 + \mathbf{v}\times\mathbf{B}_0\Big)\cdot\nabla_{\mathbf{v}} f = S(\mathbf{x},\mathbf{v},t), \tag{5.26}$$

a steady background $f_0(\mathbf{x},\mathbf{v})$ satisfies $\partial_t f_0 = 0$. The steady source S_0 that exactly sustains f_0 is therefore

$$S_0(\mathbf{x},\mathbf{v}) = -\,\mathbf{v}\cdot\nabla_{\mathbf{x}} f_0 - \frac{q}{m}\Big(\mathbf{E}_0 + \mathbf{v}\times\mathbf{B}_0\Big)\cdot\nabla_{\mathbf{v}} f_0. \tag{5.27}$$

If we take f_0 spatially uniform and isotropic,

$$f_0(\mathbf{v}) = C\,\mathrm{e}^{-m\mathbf{v}^2}, \qquad v = \|\mathbf{v}\|, \tag{5.28}$$

then $\nabla_{\mathbf{x}} f_0 = \mathbf{0}$ and $\nabla_{\mathbf{v}} f_0 = -2\,m\,\mathbf{v}\,f_0$. Since $\mathbf{v}\cdot(\mathbf{v}\times\mathbf{B}_0) = 0$, the magnetic term does not contribute, and (5.27) reduces to

$$S_0(\mathbf{v}) = -\frac{q}{m}\,\mathbf{E}_0\cdot\Big(-2\,m\,\mathbf{v}\,f_0\Big) = -2\;q\,(\mathbf{E}_0\cdot\mathbf{v})\,f_0. \tag{5.29}$$

The expression for the source term does not imply a functional dependence $S = S(E_0)$. Rather, S is a prescribed input (e.g., a particle injection rate) set by the experiment, and E_0 is the steady electric field produced in response. The equation merely relates S and E_0; it does not make S a dependent variable.

5.5 Lyapunov functional and stability framework for constant EM field

To analyse the stability of a steady Vlasov–Maxwell state, we employ a Lyapunov functional: a positive-definite quantity whose time derivative along solutions captures the balance between energy-preserving and dissipative mechanisms. If this functional is non-increasing in time, the perturbations decay, establishing Lyapunov stability.

We work with perturbations $(f', \mathbf{E}', \mathbf{B}')$ about a steady background $(f_0, \mathbf{E}_0$ $, \mathbf{B}_0)$. We first define the Lyapunov functional in this form

$$\mathcal{L}[f', \mathbf{E}', \mathbf{B}'] := \iint \frac{f'^2}{2W}\,\mathrm{d}\mathbf{v}\,\mathrm{d}\mathbf{x} \;+\; \frac{\varepsilon_0}{2}\int |\mathbf{E}'|^2\,\mathrm{d}\mathbf{x} \;+\; \frac{1}{2\mu_0}\int |\mathbf{B}'|^2\,\mathrm{d}\mathbf{x}, \tag{5.30}$$

where the weight $W(\mathbf{v})$ is chosen so that the kinetic–field coupling cancels exactly with the time derivative of the field energy, following the Poynting identity. A convenient and general choice is

$$W(\mathbf{v}) \;=\; 2\,f_0(\mathbf{v}), \tag{5.31}$$

which is positive for any background distribution decreasing with particle energy. Appendix A gives the reason why this form of Lyapunov function is selected. To study the stability of the perturbations, we consider the time derivative of the Lyapunov functional $\mathcal{L}[f', \mathbf{E}', \mathbf{B}']$. Denoting by S the source term, the time derivative reads

$$\frac{\mathrm{d}\mathcal{L}}{\mathrm{d}t} = \iint \frac{f'}{W}\,\frac{\partial f'}{\partial t}\,\mathrm{d}\mathbf{v}\,\mathrm{d}\mathbf{x} + \varepsilon_0\int \mathbf{E}'\cdot\frac{\partial \mathbf{E}'}{\partial t}\,\mathrm{d}\mathbf{x} + \frac{1}{\mu_0}\int \mathbf{E}'\cdot\frac{\partial \mathbf{E}'}{\partial t}\,\mathrm{d}\mathbf{x}. \tag{5.32}$$

the evolution of $\frac{\mathrm{d}\mathcal{L}}{\mathrm{d}t}$ naturally separates into a kinetic part driven by the f'–equation and a field part governed by Maxwell's equations for $(\mathbf{E}', \mathbf{B}')$. We begin with the full Vlasov equation for the total distribution function $f(t, \mathbf{x}, \mathbf{v})$ from (5.26) and decompose it into a steady background $f_0(\mathbf{x}, \mathbf{v})$ and a perturbation $f'(t, \mathbf{x}, \mathbf{v})$, The background fields $(f_0, \mathbf{E}_0, \mathbf{B}_0)$ satisfy the steady Vlasov equation

$$\mathbf{v}\cdot\nabla_{\mathbf{x}} f_0 + \mathbf{a}_0\cdot\nabla_{\mathbf{v}} f_0 = S_0, \quad \mathbf{a}_0 := \frac{q}{m}\left(\mathbf{E}_0 + \mathbf{v}\times\mathbf{B}_0\right), \tag{5.33}$$

From (5.26) with $f = f_0 + f'$, the left–hand side splits linearly:

$$\Big[\partial_t f_0 + \mathbf{v}\cdot\nabla_{\mathbf{x}} f_0 + (\mathbf{a}_0 + \mathbf{a}')\cdot\nabla_{\mathbf{v}} f_0\Big] + \Big[\partial_t f' + \mathbf{v}\cdot\nabla_{\mathbf{x}} f' + (\mathbf{a}_0 + \mathbf{a}')\cdot\nabla_{\mathbf{v}} f'\Big] = S_0(\mathbf{x},\mathbf{v}). \tag{5.34}$$

and subtracting the equilibrium equation for f_0, this yields the linearized perturbation equation

$$\frac{\partial f'}{\partial t} + \mathbf{v}\cdot\nabla_{\mathbf{x}} f' + \mathbf{a}_0\cdot\nabla_{\mathbf{v}} f' + \mathbf{a}'\cdot\nabla_{\mathbf{v}} f_0 = 0, \tag{5.35}$$

where $\mathbf{a}' := \frac{q}{m}(\mathbf{E}' + \mathbf{v}\times\mathbf{B}')$ represents the acceleration perturbation and the source term is canceled after subtraction.

5.5.1 Linear Stability via a Lyapunov Functional

In the last part, we constructed an expression for f' which is the basic information we need to conduct for future analysis. In our expression (5.35) we neglect second-order perturbation terms when computing $\dot{L}$ such as $\mathbf{a}'\cdot\nabla_{\mathbf{v}} f'$ since it is a product of two perturbations: the field perturbation $\mathbf{a}'$ and the distribution perturbation f', thus restricting the calculation to linearized dynamics.

This procedure is equivalent to performing a linear stability analysis in Lyapunov form: the resulting $\dot{L}$ describes the time evolution of small perturbations governed by the linearized system. Through the Lyapunov direct method, the asymptotic stability of the linearized system implies the local Lyapunov stability of the full nonlinear system under suitable smoothness conditions. Thus, although the present result is derived from the linearized equations, it remains a valuable step toward understanding the stability properties of the nonlinear Vlasov–Maxwell system.

The kinetic part that appears when integrating over phase space:

$$\mathcal{I} := \iint \frac{f'}{W}\Big(-\mathbf{v}\cdot\nabla_{\mathbf{x}} f' - \mathbf{a}_0\cdot\nabla_{\mathbf{v}} f' - \mathbf{a}'\cdot\nabla_{\mathbf{v}} f_0\Big)\,\mathrm{d}\mathbf{v}\,\mathrm{d}\mathbf{x}. \tag{5.36}$$

Integrate by parts in $\mathbf{x}$ only for the first term:

$$\begin{aligned}
-\int_{\Omega_x} \frac{f'}{W}\,\mathbf{v}\cdot\nabla_x f'\,\mathrm{d}x &= -\frac{1}{2}\int_{\Omega_x} \mathbf{v}\cdot\nabla_x\Big(\frac{f'^2}{W}\Big)\,\mathrm{d}x + \frac{1}{2}\int_{\Omega_x} f'^2\,\mathbf{v}\cdot\nabla_x\Big(\frac{1}{W}\Big)\,\mathrm{d}x \\
&= -\frac{1}{2}\int_{\Omega_x} \nabla_x\cdot\Big(\frac{f'^2}{W}\,\mathbf{v}\Big)\,\mathrm{d}x + \frac{1}{2}\int_{\Omega_x} \frac{f'^2}{W}\,\nabla_x\cdot\mathbf{v}\,\mathrm{d}x + \frac{1}{2}\int_{\Omega_x} f'^2\,\mathbf{v}\cdot\nabla_x\Big(\frac{1}{W}\Big)\,\mathrm{d}x \\
&= -\frac{1}{2}\int_{\partial\Omega_x} (\mathbf{v}\cdot\mathbf{n})\,\frac{f'^2}{W}\,\mathrm{d}S + \frac{1}{2}\int_{\Omega_x} \frac{f'^2}{W}\,\nabla_x\cdot\mathbf{v}\,\mathrm{d}x + \frac{1}{2}\int_{\Omega_x} f'^2\,\mathbf{v}\cdot\nabla_x\Big(\frac{1}{W}\Big)\,\mathrm{d}x.
\end{aligned} \tag{5.37}$$

As $\mathbf{v}$ is a velocity coordinate (not a field in x), it is independent of x, so $\nabla_x\cdot\mathbf{v} = 0$; hence the second volume term $\frac{1}{2}\int_{\Omega_x}\frac{f'^2}{W}\,\nabla_x\cdot\mathbf{v}\,\mathrm{d}x$ vanishes. W does not depend on x from our defination, then $\nabla_x(1/W) = 0$; thus the third term $\frac{1}{2}\int_{\Omega_x} f'^2\,\mathbf{v}\cdot\nabla_x(1/W)\,\mathrm{d}x$ also vanishes. Only the boundary flux term remains.

This boundary integral vanishes under the spatial BCs from Section 4. For the second term, using

$$\frac{f'}{W}\,\mathbf{a}_0\cdot\nabla_{\mathbf{v}} f' = \mathbf{a}_0\cdot\nabla_{\mathbf{v}}\Big(\frac{f'^2}{2W}\Big) + \frac{f'^2}{2}\,\nabla_{\mathbf{v}}\cdot\Big(\frac{\mathbf{a}_0}{W}\Big), \tag{5.38}$$

integrating over $\mathbf{v}$ on the spherical velocity space and then letting $R \to \infty$, we obtain

$$\int_{\mathbb{R}^3} \frac{f'}{W}\, \mathbf{a}_0 \cdot \nabla_{\mathbf{v}} f' \, \mathrm{d}\mathbf{v} = \frac{1}{2} \int_{|\mathbf{v}|=R} \left(\frac{f'^2}{W}\right) \mathbf{a}_0 \cdot \mathbf{n}_{\mathbf{v}} \, \mathrm{d}S_{\mathbf{v}} \; + \; \frac{1}{2} \int_{\mathbb{R}^3} f'^2 \, \nabla_{\mathbf{v}} \cdot \left(\frac{\mathbf{a}_0}{W}\right) \mathrm{d}\mathbf{v}. \tag{5.39}$$

The velocity–sphere flux (first term) vanishes under the BCs of the velocity space from Section 4. For the bulk term,

$$\nabla_{\mathbf{v}} \cdot \left(\frac{\mathbf{a}_0}{W}\right) = \frac{1}{W} \nabla_{\mathbf{v}} \cdot \mathbf{a}_{0+} \mathbf{a}_0 \cdot \nabla_{\mathbf{v}} \left(\frac{1}{W}\right) = \mathbf{a}_0 \cdot \nabla_{\mathbf{v}} \left(\frac{1}{W}\right), \tag{5.40}$$

since $\nabla_{\mathbf{v}} \cdot (\mathbf{v} \times \mathbf{B}_0) = 0$ and E_0 is independent of v , the first term gives zero. This term can be further derived into

$$\nabla_{\mathbf{v}} \cdot \left(\frac{\mathbf{a}_0}{W}\right) = \mathbf{a}_0 \cdot \nabla_{\mathbf{v}} \left(\frac{1}{2\, f_0}\right) = \frac{\mathbf{a}_0}{2} (-1) \frac{(2\, m\, v)}{f_0}, \tag{5.41}$$

Substituting the result of S_0 from (5.29) into produces the bulk transport term

$$\mathcal{I}_{\mathrm{tr}} = \frac{1}{2m} \iint f'^2 \, \frac{S_0}{f_0^2} \, \mathrm{d}\mathbf{x} \, \mathrm{d}\mathbf{v}. \tag{5.42}$$

For the coupling term $\iint \frac{f'}{f_0}\, a' \cdot \nabla_v f_0 \, dx \, dv$, the linearized Vlasov equation the perturbation of the Lorentz force is

$$a'(x, v, t) = \frac{q}{m} \big(E'(x, t) + v \times B'(x, t)\big). \tag{5.43}$$

We obtained the equilibrium distribution in (5.13) so that

$$\nabla_v f_0 = f_0 \, (-2mv). \tag{5.44}$$

Therefore

$$\begin{aligned} \iint \frac{f'}{2f_0}\, a' \cdot \nabla_{\mathbf{v}} f_0 \, \mathrm{d}\mathbf{x} \, \mathrm{d}\mathbf{v} &= \iint \frac{f'}{2f_0} \, \frac{q}{m} \big(\mathbf{E}' + \mathbf{v} \times \mathbf{B}'\big) \cdot \big(f_0(-2m\mathbf{v})\big) \, \mathrm{d}\mathbf{x} \, \mathrm{d}\mathbf{v} \\ &= q \iint f'(-1) \left(\mathbf{E}' \cdot \mathbf{v} + (\mathbf{v} \times \mathbf{B}') \cdot \mathbf{v}\right) \mathrm{d}\mathbf{x} \, \mathrm{d}\mathbf{v} \\ &= -\, q \iint f' \, (\mathbf{E}' \cdot \mathbf{v}) \, \mathrm{d}\mathbf{x} \, \mathrm{d}\mathbf{v}, \end{aligned} \tag{5.45}$$

because $(v \times B') \cdot v = 0$ identically. It is convenient to pull out E' and define the current perturbation

$$\mathbf{J}'(\mathbf{x}, t) := q \int_{\mathbb{R}^3} \mathbf{v} \, f'(\mathbf{x}, \mathbf{v}, t) \, \mathrm{d}\mathbf{v}, \tag{5.46}$$

so that (5.45) becomes

$$\iint \frac{f'}{f_0}\, \mathbf{a}' \cdot \nabla_{\mathbf{v}} f_0 \, \mathrm{d}\mathbf{x} \, \mathrm{d}\mathbf{v} \; = \; -\int_{\Omega_x} \mathbf{E}'(\mathbf{x}, t) \cdot \mathbf{J}'(\mathbf{x}, t) \, \mathrm{d}\mathbf{x}, \tag{5.47}$$

No integration by parts is used here, so no boundary condition is needed beyond integrability ensuring that J' exists.

The last two term in (5.32) require the expression of $\mathbf{E}'$, $\mathbf{B}'$ and their time derivative. From (5.5), we

can rewrite the Maxwell's equations into the following expression

$$\nabla\cdot(\mathbf{E}_0+\mathbf{E}') = \frac{\rho}{\varepsilon_0}, \tag{5.48}$$

$$\nabla\cdot(\mathbf{B}_0+\mathbf{B}') = 0, \tag{5.49}$$

$$\nabla\times(\mathbf{E}_0+\mathbf{E}') = -\Big(\cancelto{0}{\partial_t\mathbf{B}_0}+\partial_t\mathbf{B}'\Big) = -\,\partial_t\mathbf{B}', \tag{5.50}$$

$$\nabla\times(\mathbf{B}_0+\mathbf{B}') = \mu_0(\mathbf{J}_0+\mathbf{J}')+\mu_0\varepsilon_0\Big(\cancelto{0}{\partial_t\mathbf{E}_0}+\partial_t\mathbf{E}'\Big) = \mu_0(\mathbf{J}_0+\mathbf{J}')+\mu_0\varepsilon_0\,\partial_t\mathbf{E}'. \tag{5.51}$$

From the boundary condition we suggested in Section 4 and subtract it by the expression (5.7), we can rewrite the last two Maxwell's equations for the perturbation fields into

$$\nabla\times\mathbf{B}' = \mu_0\varepsilon_0\,\partial_t\mathbf{E}'+\mu_0\,\mathbf{J}', \tag{5.52}$$

$$\nabla\times\mathbf{E}' = -\,\partial_t\mathbf{B}'. \tag{5.53}$$

Recognising that from formula (5.32) and the Maxwell's equations for the perturbation fields ,we arrive at the perturbation Poynting theorem:

$$\begin{aligned}
\varepsilon_0\int_{\Omega_x}\mathbf{E}'\cdot\partial_t\mathbf{E}'\,\mathrm{d}\mathbf{x}+\frac{1}{\mu_0}\int_{\Omega_x}\mathbf{B}'\cdot\partial_t\mathbf{B}'\,\mathrm{d}\mathbf{x} &= \int_{\Omega_x}\left[\frac{1}{\mu_0}\mathbf{E}'\cdot(\nabla\times\mathbf{B}')-\mathbf{E}'\cdot\mathbf{J}'-\frac{1}{\mu_0}\mathbf{B}'\cdot(\nabla\times\mathbf{E}')\right]\mathrm{d}\mathbf{x}\\
&= -\int_{\Omega_x}\mathbf{E}'\cdot\mathbf{J}'\,\mathrm{d}\mathbf{x}+\frac{1}{\mu_0}\int_{\Omega_x}\big[\mathbf{E}'\cdot(\nabla\times\mathbf{B}')-\mathbf{B}'\cdot(\nabla\times\mathbf{E}')\big]\mathrm{d}\mathbf{x}\\
&= -\int_{\Omega_x}\mathbf{E}'\cdot\mathbf{J}'\,\mathrm{d}\mathbf{x}-\frac{1}{\mu_0}\int_{\Omega_x}\nabla\cdot(\mathbf{E}'\times\mathbf{B}')\,\mathrm{d}\mathbf{x}\\
&= -\int_{\Omega_x}\mathbf{E}'\cdot\mathbf{J}'\,\mathrm{d}\mathbf{x}-\frac{1}{\mu_0}\int_{\partial\Omega_x}(\mathbf{E}'\times\mathbf{B}')\cdot\mathbf{n}\,\mathrm{d}S.
\end{aligned} \tag{5.54}$$

The surface term in (5.54) is eliminated under the Electromagnetic BCs from Section 4. ensures

$$(\mathbf{E}'\times\mathbf{B}')\cdot\mathbf{n} = (\mathbf{n}\times\mathbf{E}')\cdot\mathbf{B}' = 0, \tag{5.55}$$

so no electromagnetic energy leaves or enters the domain. With these conditions, (5.54) reduces to

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[\frac{\varepsilon_0}{2}\int|\mathbf{E}'|^2\,\mathrm{d}\mathbf{x}+\frac{1}{2\mu_0}\int|\mathbf{B}'|^2\,\mathrm{d}\mathbf{x}\right] = -\int_{\Omega_x}\mathbf{E}'\cdot\mathbf{J}'\,\mathrm{d}\mathbf{x}, \tag{5.56}$$

which expresses that the field perturbation energy changes solely via work done on the particles.

After the cancelations on Field term and acceleration pertubation term, the time derivative of the Lyapunov functional simplifies to

$$\frac{\mathrm{d}}{\mathrm{d}t}\,\mathcal{L}[f',\mathbf{E}',\mathbf{B}'] = -\frac{1}{2m}\iint f'^2\,\frac{S_0}{f_0^2}\,\mathrm{d}\mathbf{x}\,\mathrm{d}\mathbf{v}, \tag{5.57}$$

In our configuration, the source term represents the injection of particles into the domain. As the exit acts as an absorbing sink, a negative source would also remove particles inside the domain as well; combined with the sink, this would rapidly drain the system and make the model trivial (nearly vacuum).

Therefore, we assume that the source is nonnegative (allowing zero) and handle particle removal only through the exit sink. Since the source term S_0 is non-negative, the integrand is always positive, and the right-hand side is therefore non-positive. This shows that the Lyapunov functional $\mathcal{L}$ is monotonically decreasing in time. In other words, the system is linearly Lyapunov stable, as the functional provides a positive quadratic measure of perturbation size that dissipates under the dynamics. While here we have established stability at the linear level, this framework also lays the foundation for the nonlinear Lyapunov analysis, which requires retaining higher-order perturbation terms.

5.6 Lyapunov Stability with Second–order perturbation

Beyond linear theory, keeping the quadratic product of perturbations produces an extra term in the kinetic part of the Lyapunov balance. In writing $\mathbf{a}' := \frac{q}{m}\big(\mathbf{E}' + \mathbf{v} \times \mathbf{B}'\big)$, the non-linear transport contribution is

$$T_{\mathrm{nl}} := -\iint \frac{f'}{W}\, \mathbf{a}' \cdot \nabla_{\mathbf{v}} f' \, \mathrm{d}\mathbf{x}\, \mathrm{d}\mathbf{v}, \tag{5.58}$$

which is a product of two perturbations (distribution f' and field increment $\mathbf{a}'$), hence, second order in the perturbation amplitude.
Let $\phi := f'^2/(2W)$. Using the product rule,

$$\frac{f'}{W}\, \mathbf{a}' \cdot \nabla_{\mathbf{v}} f' \;=\; \mathbf{a}' \cdot \nabla_{\mathbf{v}} \phi \;-\; \frac{f'^2}{2}\, \mathbf{a}' \cdot \nabla_{\mathbf{v}} \Big(\frac{1}{W}\Big). \tag{5.59}$$

Integrating the first term over the velocity, we get the expression below

$$-\int_{\mathbb{R}^3} \mathbf{a}' \cdot \nabla_{\mathbf{v}} \phi \, \mathrm{d}\mathbf{v} = -\int_{\mathbb{R}^3} \nabla_{\mathbf{v}} \cdot \big(\mathbf{a}'\, \phi\big)\, \mathrm{d}\mathbf{v} + \int_{\mathbb{R}^3} \phi\, \nabla_{\mathbf{v}} \cdot \mathbf{a}' \, \mathrm{d}\mathbf{v} \;=\; 0. \tag{5.60}$$

and use $\nabla_{\mathbf{v}} \cdot \mathbf{a}' = 0$ (since $\mathbf{E}'$ is independent of $\mathbf{v}$ and $\mathbf{v} \mapsto \mathbf{v} \times \mathbf{B}'$ has zero divergence) together with the decay of velocity-sphere so that the boundary integral in $|\mathbf{v}| = R$ vanishes as $R \to \infty$ from the boundary condition. Hence,

$$T_{\mathrm{nl}} \;=\; \frac{1}{2} \iint f'^2\, \mathbf{a}' \cdot \nabla_{\mathbf{v}} \Big(\frac{1}{W}\Big)\, \mathrm{d}\mathbf{x}\, \mathrm{d}\mathbf{v}. \tag{5.61}$$

For a Maxwellian background $f_0(\mathbf{v}) = C\, \mathrm{e}^{-m|\mathbf{v}|^2}$ with $W = 2\, f_0$, we have

$$\nabla_{\mathbf{v}} \Big(\frac{1}{W}\Big) = \nabla_{\mathbf{v}} \Big(\frac{1}{2 f_0}\Big) = \frac{m\, \mathbf{v}}{f_0}. \tag{5.62}$$

Therefore,

$$T_{\mathrm{nl}} = \frac{1}{2} \iint \frac{f'^2}{f_0}\, \mathbf{a}' \cdot \mathbf{v} \; \mathrm{d}\mathbf{x}\, \mathrm{d}\mathbf{v} = \frac{q}{2m} \iint \frac{f'^2}{f_0}\, \mathbf{E}' \cdot \mathbf{v} \; \mathrm{d}\mathbf{x}\, \mathrm{d}\mathbf{v}, \tag{5.63}$$

since $\mathbf{v} \cdot (\mathbf{v} \times \mathbf{B}') = 0$. After this we can substitute the result into (5.57) and obtain the final expression in the form of

$$\frac{\mathrm{d}}{\mathrm{d}t}\, \mathcal{L}[f', \mathbf{E}', \mathbf{B}'] = -\frac{1}{2} \iint f'^2\, \frac{S_0}{f_0^2}\, \mathrm{d}\mathbf{x}\, \mathrm{d}\mathbf{v} + \frac{q}{2m} \iint \frac{f'^2}{f_0}\, \mathbf{E}' \cdot \mathbf{v} \; \mathrm{d}\mathbf{x}\, \mathrm{d}\mathbf{v}, \tag{5.64}$$

The first term is non-positive from our analysis in the last section. The second term is sign–indefinite

because $\mathbf{E}' \cdot \mathbf{v}$ changes sign across phase space; hence we cannot conclude $\dot{\mathcal{L}} \leq 0$ globally. To address this, we can apply the background–flow (auxiliary functional) method in the future a carefully chosen coupling term so that, after differentiation and using Maxwell's equations/boundary conditions, the indefinite volume integral is cancelled or absorbed into negative–definite field–energy/flux terms, yielding $\frac{\mathrm{d}}{\mathrm{d}t}\,\widetilde{\mathcal{L}} \leq 0$.

6 Extension from constant to x–dependent fields

To build a more realistic model of the Vlasov–Maxwell system, it is necessary to give both the electric and magnetic fields additional degrees of freedom. In the earlier model, the $\mathbf{B}$-field was taken as uniform constants. Here we extend this framework by allowing spatial dependence in $\mathbf{x}$, while still retaining an exponential stationary distribution that is consistent with the steady Vlasov equation. This approach follows the construction in Markov [19], where stationary distributions are designed to remain constant along particle trajectories in phase space.

6.1 Additional boundary conditions for the potential-based model

In extending the model to allow spatially varying fields, we introduce scalar potentials $\phi(\mathbf{x})$ and $\psi(\mathbf{x})$ together with a drift direction $d(\mathbf{x})$. These functions appear in the exponential ansatz for the stationary distribution and must therefore satisfy additional boundary conditions to ensure consistency with Maxwell's equations and invariants of motion.

Orthogonality conditions. The first set of constraints enforces that ϕ and ψ do not vary along the drift direction d:

$$d \cdot \nabla\phi = 0, \qquad d \cdot \nabla\psi = 0. \tag{6.1}$$

Here d is the fixed drift vector, while ϕ and ψ are effective scalar potentials associated with the electric and magnetic fields, respectively.

Scalar potential boundary data. The electric field is represented as $E = -\nabla U$, where U is the scalar potential. On different parts of the boundary $\partial\Omega$ we impose

$$U|_{\Gamma_{\text{metal}}} = U_0, \qquad [\![U]\!]_{\Gamma_{\text{diel}}} = 0, \qquad n \cdot \nabla U|_{\Gamma_{\text{open}}} \approx 0, \tag{6.2}$$

corresponding respectively to conducting walls, dielectric interfaces, and open boundaries. These conditions are the natural extensions of the ones already discussed in Section 4.

Vector potential boundary data. For the magnetic field $B = \nabla \times A$, only the projection of A along the drift d needs to be prescribed:

$$(A \cdot d)\big|_{\partial\Omega_x} = u_1(x). \tag{6.3}$$

This condition fixes the magnetic contribution consistent with the chosen ansatz.

Together, (6.1)–(6.3) provide the additional boundary information required by the potential-based formulation of the steady Vlasov–Maxwell system.

6.2 Solution for f_0

A broad class of steady distributions can be written as follows

$$f(\mathbf{x}, \mathbf{v}) = f\big(-\alpha|\mathbf{v}|^2 + \varphi(\mathbf{x}), \;\; \mathbf{v} \cdot \mathbf{d}(\mathbf{x}) + \psi(\mathbf{x})\big)\,, \tag{6.4}$$

where $\alpha > 0$ is a constant, while $\varphi(\mathbf{x})$, $\psi(\mathbf{x})$, and the vector field $\mathbf{d}(\mathbf{x})$ are functions determined by the equilibrium. The variables φ and ψ act as effective scalar potentials associated with the electric and magnetic parts of the equilibrium, respectively, and $\mathbf{d}$ plays the role of a drift direction linking velocity and spatial dependence. In the simplest setting $\mathbf{d}$ is constant, reducing (6.4) to the constant-field case.

A particularly convenient choice within this class is the exponential (Gaussian) ansatz

$$f(\mathbf{x}, \mathbf{v}) = C \exp\Big(- \alpha|\mathbf{v}|^2 + \phi(\mathbf{x}) + \mathbf{v} \cdot d(\mathbf{x}) + \psi(\mathbf{x})\Big), \qquad \alpha > 0, \tag{6.5}$$

where C is a normalisation constant. This reduces to a shifted Maxwellian when $d \neq 0$, thereby allowing equilibrium bulk flow. Compared with the purely constant-field model, (6.5) introduces spatially dependent potentials ϕ and ψ together with a drift direction d, enabling $\mathbf{E}$ and $\mathbf{B}$ to vary consistently in space.

6.3 Steady background solution for E_0 and B_0

We now derive explicit expressions for the steady electromagnetic fields consistent with the chosen ansatz for the distribution function in the presence of sources. Starting from the steady Vlasov equation,

$$\mathbf{v} \cdot \nabla_x f_0 + \frac{q}{m}\left(E_0 + \frac{1}{c}\mathbf{v} \times B_0\right) \cdot \nabla_v f_0 \;=\; S_0(\mathbf{x}, \mathbf{v}), \tag{6.6}$$

we assume that the source $S_0(\mathbf{x}, \mathbf{v})$ admits a general expansion in powers of v:

$$S_0(\mathbf{x}, \mathbf{v}) = S_c(\mathbf{x}) + \mathbf{v} \cdot \big(s_E(\mathbf{x}) + s_B(\mathbf{x})\big) + \mathbf{v}^T Q(\mathbf{x})\mathbf{v}, \tag{6.7}$$

where $S_c(\mathbf{x})$ is a constant-in-v part, the linear term has been split into electric and magnetic channels $s_E(\mathbf{x}), s_B(\mathbf{x})$, and $Q(\mathbf{x})$ is a placeholder for possible quadratic contributions. In this section, we retain the linear part explicitly and treat the others as placeholders to be developed later.

Action of derivatives. Computing derivatives directly on the ansatz (6.5) gives

$$\nabla_x f_0 = f_0 \big(\nabla\phi(\mathbf{x}) + \nabla\psi(\mathbf{x})\big), \tag{6.8}$$

$$\nabla_v f_0 = f_0 \big(- 2\alpha\mathbf{v} + d(\mathbf{x})\big). \tag{6.9}$$

Substitution. Substituting into the Vlasov equation

$$\begin{aligned} \mathbf{v} \cdot\; f_0(\nabla\phi + \nabla\psi) + \frac{q}{m}\Big(E + \frac{1}{c}\,\mathbf{v} \times B\Big) \cdot\; f_0(-2\alpha\mathbf{v} + d) \\ = S_c(\mathbf{x}) + \mathbf{v} \cdot \big(s_E(\mathbf{x}) + s_B(\mathbf{x})\big) + \mathbf{v}^T Q(\mathbf{x}). \end{aligned} \tag{6.10}$$

Collecting terms. Grouping by powers of v and imposing the orthogonality conditions in Section 6.1 the quadratic contribution is killed . The linear coefficient in v then gives the balance condition

$$\nabla(\phi + \psi) - \frac{2\alpha q}{m}E + \frac{q}{mc}B \times d \;=\; \frac{s_E + s_B}{f_0}, \tag{6.11}$$

while the constant and quadratic parts in (6.7) remain to be accounted for in later analysis.

Splitting by channels. Equation (6.11) can be decomposed into two separate balance conditions:

$$\frac{2\alpha q}{m} E = \nabla\phi + \frac{-s_E}{f_0}, \tag{6.12}$$

$$\frac{q}{mc} B \times d = -\nabla\psi + \frac{s_B}{f_0}. \tag{6.13}$$

Explicit expressions. We therefore obtain the background steady fields

$$E_0(\mathbf{x}) = \frac{m}{2\alpha q}\big(\nabla\phi - s_E\big), \tag{6.14}$$

$$B_0(\mathbf{x}) \times d = -\frac{mc}{q}\big(\nabla\psi - s_B\big). \tag{6.15}$$

The expression for E_0 shows that the equilibrium electric field is governed by the gradient of ϕ, corrected by the electric source s_E which represents external forcing. Similarly, (6.15) constrains the magnetic field through a curl-type relation involving the potential ψ and the magnetic source s_B.

To obtain an explicit form for B_0, we cross both sides of (6.15) with d. Using the vector triple-product identity

$$d \times (B \times d) \;=\; (d \cdot d)\, B - (B \cdot d)\, d, \tag{6.16}$$

we obtain

$$|d|^2 B - (B \cdot d)\, d \;=\; -\frac{mc}{q}\, d \times (\nabla\psi - s_B). \tag{6.17}$$

Defining $\lambda(\mathbf{x}) := B \cdot d$, the component of B parallel to d, this yields

$$B(x) \;=\; \frac{\lambda(\mathbf{x})}{|d|^2}\, d \;-\; \frac{mc}{q|d|^2}\, d \times \big(\nabla\psi - s_B\big). \tag{6.18}$$

Equation (6.18) shows that B is uniquely fixed in the subspace perpendicular to d by the Vlasov balance, while its parallel component, encoded in $\lambda(\mathbf{x})$, remains arbitrary. This freedom is subsequently constrained by Maxwell's condition $\nabla \cdot B = 0$ and by boundary data. In simplified models, one may impose the gauge condition $B \cdot d = 0$, in which case $\lambda(\mathbf{x}) \equiv 0$ and the magnetic field reduces to

$$B_0(\mathbf{x}) \;=\; -\frac{mc}{q|d|^2}\, d \times \big(\nabla\psi - s_B\big), \tag{6.19}$$

which is manifestly perpendicular to d.

6.4 solution for linear and quadratic term of S_0

Polynomial structure in v. After substituting f into the Vlasov equation and dividing through by f, the left-hand side becomes a polynomial in v:

$$\mathbf{v}^{\mathsf{T}}\big(\tfrac{1}{2}\big(\nabla d + (\nabla d)^{\mathsf{T}}\big)\,\mathbf{v} + \mathbf{v} \cdot \Big(\nabla(\phi + \psi) - \tfrac{2\alpha q}{m} E_0 + \tfrac{q}{mc} B_0 \times d\Big) + \frac{q}{m} E_0 \cdot d = S_0(\mathbf{x}, \mathbf{v}), \tag{6.20}$$

where the first term is quadratic in $\mathbf{v}$, the second is linear, and the last is constant.

Matching coefficients. By construction, the linear coefficient equals $s_E + s_B$, so the source must be

$$S_0(\mathbf{x},\mathbf{v}) \;=\; \Big[\mathbf{v}\cdot(s_E+s_B) + \frac{q}{m}E_0\cdot d + \mathbf{v}^{\mathsf{T}}\big(\tfrac{1}{2}\big(\nabla d + (\nabla d)^{\mathsf{T}}\big)\,\mathbf{v}\Big]. \tag{6.21}$$

Quadratic and constant pieces. From (6.21) we can isolate the higher-order contributions:

$$S^{(2)}(\mathbf{x},\mathbf{v}) = \;\mathbf{v}^{\mathsf{T}}\Big(\tfrac{1}{2}\big(\nabla d + (\nabla d)^{\mathsf{T}}\big)\Big)\,\mathbf{v}, \tag{6.22}$$

$$S^{(0)}(\mathbf{x}) = \;\frac{q}{m}E_0\cdot d \;=\; \frac{1}{2\alpha}\Big[(\nabla\phi)\cdot d - (s_E\cdot d)\Big]. \tag{6.23}$$

$(\nabla\phi)\cdot d = 0$ which is based on the boundary condition we have set for this system which can rewrite (6.23) into

$$S^{(0)}(\mathbf{x}) = \;\frac{q}{m}E_0\cdot d \;=\; \frac{1}{2\alpha}\Big[-(s_E\cdot d)\Big]. \tag{6.24}$$

Remarks.

- If d is constant, the quadratic contribution $S^{(2)}$ vanishes.
- The magnetic field B_0 does not contribute to the quadratic or constant terms; it affects only the linear balance through s_B.

After this rearrange we can rewrite the source term into

$$S_0(\mathbf{x},\mathbf{v}) \;=\; \Big[\mathbf{v}\cdot(s_E(\mathbf{x})+s_B(\mathbf{x})) + \frac{1}{2\alpha}\Big[-(s_E\cdot d)\Big] + \mathbf{v}^{\mathsf{T}}\big(\tfrac{1}{2}\big(\nabla d + (\nabla d)^{\mathsf{T}}\big)\,\mathbf{v}\Big]. \tag{6.25}$$

6.5 Steady–state current density

We now compute the charge current density associated with the exponential ansatz for the steady distribution function,

$$f(\mathbf{x},\mathbf{v}) \;=\; C\exp\Big(-\alpha|\mathbf{v}|^2 + \phi(\mathbf{x}) + v\cdot d(\mathbf{x}) + \psi(\mathbf{x})\Big), \qquad \alpha > 0. \tag{6.26}$$

The linear term $v\cdot d$ can be combined with $-\alpha|v|^2$ by completing the square:

$$-\alpha|\mathbf{v}|^2 + \mathbf{v}\cdot d \;=\; -\alpha\Big|\mathbf{v} - \tfrac{d}{2\alpha}\Big|^2 + \frac{|d|^2}{4\alpha}. \tag{6.27}$$

This motivates the definitions

$$u(\mathbf{x}) \;:=\; \frac{d(\mathbf{x})}{2\alpha}, \qquad A(\mathbf{x}) \;:=\; C\exp\Big(\phi(\mathbf{x}) + \psi(\mathbf{x}) + \tfrac{|d(\mathbf{x})|^2}{4\alpha}\Big), \tag{6.28}$$

so that the distribution takes the Gaussian form

$$f(\mathbf{x},\mathbf{v}) \;=\; A(\mathbf{x})\,\exp\big(-\alpha|\mathbf{v}-u(\mathbf{x})|^2\big). \tag{6.29}$$

Thus f is a shifted Maxwellian with mean drift velocity $u(\mathbf{x})$ and amplitude $A(\mathbf{x})$.

Zeroth moment (density). The number density is obtained as the zeroth moment,

$$n(\mathbf{x}) \;=\; \int_{\mathbb{R}^3} f(\mathbf{x},\mathbf{v})\,\mathrm{d}v \;=\; A(\mathbf{x}) \int_{\mathbb{R}^3} e^{-\alpha|\mathbf{v}-u|^2}\,\mathrm{d}v \;=\; A(\mathbf{x})\left(\frac{\pi}{\alpha}\right)^{3/2}. \tag{6.30}$$

First moment (flux). The particle flux follows from the first velocity moment,

$$\Gamma(\mathbf{x}) \;=\; \int_{\mathbb{R}^3} \mathbf{v} f(\mathbf{x},\mathbf{v})\,\mathrm{d}v \;=\; u(\mathbf{x})\,n(\mathbf{x}), \tag{6.31}$$

so that the mean drift velocity $u(\mathbf{x})$ is the average velocity of the distribution.

Current density. Multiplying by the charge q gives the steady–state charge current density,

$$J(\mathbf{x}) \;=\; q\Gamma(\mathbf{x}) \;=\; q\,n(\mathbf{x})\,u(\mathbf{x}) \;=\; \frac{q}{2\alpha}\,n(\mathbf{x})\,d(\mathbf{x}). \tag{6.32}$$

In explicit parameters of the ansatz (6.26), this becomes

$$J(\mathbf{x}) \;=\; q\Big(\frac{\pi}{\alpha}\Big)^{3/2} C \exp\Big(\phi(\mathbf{x}) + \psi(\mathbf{x}) + \frac{|d(\mathbf{x})|^2}{4\alpha}\Big)\,\frac{d(\mathbf{x})}{2\alpha}. \tag{6.33}$$

Interpretation. Equation (6.32) shows that the steady current is proportional to the equilibrium density $n(\mathbf{x})$ and aligned with the drift vector $d(\mathbf{x})$. The factor $1/(2\alpha)$ reflects the role of thermal spread in reducing the net drift, while the exponential prefactor in (6.33) encodes the contributions of the scalar potentials ϕ, ψ, and the drift penalty $|d|^2/(4\alpha)$.

7 Conclusion

This study traces a clear route from the building of tractable steady states of the Vlasov–Maxwell system to the evaluation of their stability and the establishment of the machinery for quantitative bounds via the background–flow (auxiliary functional) method. The key outcomes are the following.

- **Baseline constant–field steady state.** We constructed an explicit, non-trivial equilibrium for the distribution, fields, and the sustaining source in a uniform-field setting. In doing so, we made explicit the linkage between the sustaining source and the steady electric field: for the chosen background, the required source is uniquely determined by (and consistent with) the steady Vlasov balance and the selected electric field profile. By symmetry of the background (drift-free Maxwellian), the net steady current in this baseline model is zero. This provides a clean operating point on which later perturbation analyses can be anchored.

- **Linearized Lyapunov stability.** We defined an energy-like Lyapunov functional for perturbations and computed its time derivative using the linearized Vlasov–Maxwell equations and the stated boundary/source sign conditions. In the small-perturbation (linear) regime, this derivative is nonincreasing, so perturbations cannot grow, establishing Lyapunov stability of the baseline equilibrium. When the full (nonlinear) couplings are retained, an additional work-like term appears, and the sign of the derivative is no longer determined. This identifies the obstruction to a direct nonlinear proof and motivates augmenting the functional (e.g., via a background-flow cross-term) to neutralize the indefinite contribution in future work.

- **x–dependent steady states.** We generalize the construction to spatially varying configurations by assuming a steady distribution that is tied to spatial potentials, a prescribed drift, and includes up-to-quadratic dependence on particle velocity. In parallel, we posited a sustaining source consistent with this structure (with constant and velocity-dependent channels). Under these assumptions, we obtained explicit relationships that map the chosen potentials to the steady electric and magnetic fields, and—via the drift—yield a closed-form expression for the steady current. This makes transparent how the modeling choices (potentials and drift profiles) set both the fields and the current, while organizing the source into constant, linear, and—when the drift varies—quadratic parts in velocity.Therefore, we can apply the Lyapunov stability analysis here in the future to determine whether this steady solution will be a stable solution.

Taken together, these results deliver (i) concrete equilibria suitable for linearization and benchmarking, (ii) a rigorous stability guarantee in the baseline case, and (iii) a constructive framework that connects microscopic modeling choices to macroscopic observables such as discharge current. The immediate next step is to embed an appropriate background–flow cross term into the Lyapunov functional so that the sign-indefinite coupling is neutralized and the established stability structure can be converted into computable bounds on current (and, by proxy, thrust) in Hall-thruster–like devices.

8 Further Work

We outline three focused next steps.

(i) **Background–flow augmentation and current bounds (constant–field baseline).** Building on the Lyapunov framework, the next step is to introduce a background–flow augmentation of the quadratic functional. The idea is to add a cross term of the form

$$\widetilde{L}[f', \mathbf{E}', \mathbf{B}'] = L[f', \mathbf{E}', \mathbf{B}'] + \iint \eta(x)\, \frac{f'}{f_0}\, (\mathbf{E}' \cdot \mathbf{v})\, \mathrm{d}\mathbf{v}\, \mathrm{d}\mathbf{x},$$

where $\eta(x)$ is chosen so that, after differentiation and substitution of Maxwell's equations, any sign–indefinite coupling between the perturbation distribution and the electric field is cancelled or absorbed into negative–definite contributions. The target inequality is then

$$\frac{d}{dt}\, \widetilde{L} \;\leq\; -\, c\, \mathcal{D},$$

for some dissipation functional $\mathcal{D}$ and constant $c > 0$. Integrating in time and invoking Ampère's identity,

$$\int \mathbf{E}' \cdot \mathbf{J}'\, \mathrm{d}\mathbf{x} = -\, \frac{d}{dt} \left(\tfrac{\varepsilon_0}{2} \int |\mathbf{E}'|^2\, \mathrm{d}\mathbf{x} - \tfrac{1}{2\mu_0} \int |\mathbf{B}'|^2\, \mathrm{d}\mathbf{x} \right) - \frac{1}{\mu_0} \int_{\partial\Omega} (\mathbf{E}' \times \mathbf{B}') \cdot \mathbf{n}\, \mathrm{d}\mathbf{S},$$

with $\mathbf{J}' = q \int \mathbf{v} f'\, dv$, then links the bound on $\widetilde{L}$ directly to bounds on the instantaneous and time–averaged current (or equivalently, the exit current flux).

Remark on the x–dependent background. In the constant–field baseline we imposed

$$\nabla \times \mathbf{B}_0 = 0, \qquad \partial_t \mathbf{E}_0 = \partial_t \mathbf{B}_0 = 0,$$

so $\mathbf{J}_0 = 0$ and the perturbation field energy followed the standard Poynting identity with source $-\int \mathbf{E}' \cdot \mathbf{J}'\, \mathrm{d}\mathbf{x}$. For an x–dependent background, however,

$$\nabla \times \mathbf{B}_0 = \mu_0 \mathbf{J}_0 \neq 0 \quad (\partial_t \mathbf{E}_0 = 0),$$

so there is a nonzero steady current. After subtracting the steady background, Maxwell's equations for $(\mathbf{E}', \mathbf{B}')$ keep their form, but the *Lyapunov* calculation changes via the Vlasov part:

$$\partial_t f' + \mathbf{v} \cdot \nabla_x f' + \frac{q}{m}\, (\mathbf{E}_0 + \mathbf{v} \times \mathbf{B}_0) \cdot \nabla_v f' + \frac{q}{m}\, (\mathbf{E}' + \mathbf{v} \times \mathbf{B}') \cdot \nabla_v f_0 = S'.$$

As a result, $\frac{d}{dt} L$ acquires extra bulk terms (e.g. $\frac{1}{2} \iint f'^2\, \mathbf{v} \cdot \nabla_x (1/W)\, \mathrm{d}\mathbf{v}\, \mathrm{d}\mathbf{x}$ and couplings from $(\mathbf{E}' + \mathbf{v} \times \mathbf{B}') \cdot \nabla_v f_0$) that were absent when $\nabla \times \mathbf{B}_0 = 0$ and W was x–independent. These terms require a new stability analysis (with $W = W(x, v)$ and/or a background–flow cross term) for the x–dependent case.

(ii) **Current bounds from the x–dependent Lyapunov theory.** Combine the x–dependent esti-

mate for $\frac{d}{dt}L_x$ with Maxwell's power balance to bound

$$\int \mathbf{E}' \cdot \mathbf{J}' \, dx, \qquad \Big\langle \int_{\partial\Omega_{\mathrm{out}}} (\mathbf{v}\cdot\mathbf{n})_+ \, f \, \mathrm{d}\mathbf{v} \, \mathrm{d}\mathbf{S} \Big\rangle,$$

and hence obtain (i) instantaneous estimates $\|\mathbf{J}\|$ in terms of $\|S_0\|$, $\|\nabla_x \log f_0\|$, and domain length, and (ii) time–averaged exit–current bounds after integrating in time. When needed, introduce a background–flow cross term adapted to $W(x, v)$ to neutralise the remaining sign–indefinite coupling and sharpen the current bound.

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Appendices

Appendix A: Justification of the Lyapunov Functional Choice

In the stability analysis, the Lyapunov functional must serve as a positive measure of perturbation size about a given steady state $(f_0, \mathbf{E}_0, \mathbf{B}_0)$. We adopt the form

$$\mathcal{L}[f', \mathbf{E}', \mathbf{B}'] := \iint \frac{f'^2}{W} \, \mathrm{d}\mathbf{v} \, \mathrm{d}\mathbf{x} \; + \; \varepsilon_0 \int |\mathbf{E}'|^2 \, \mathrm{d}\mathbf{x} \; + \; \frac{1}{\mu_0} \int |\mathbf{B}'|^2 \, \mathrm{d}\mathbf{x}, \tag{.1}$$

where $f' = f - f_0$ and $W(\mathbf{v}) > 0$ is a velocity weight determined by the background distribution f_0.

A.1 Positivity Requirement

The kinetic energy

$$K[f] := \iint \frac{1}{2} m |\mathbf{v}|^2 f \, \mathrm{d}\mathbf{v} \, \mathrm{d}\mathbf{x} \tag{.2}$$

is positive for $f \geq 0$. However, the difference $K[f_0 + f'] - K[f_0]$ is not necessarily positive definite:

$$\Delta K = \iint \frac{1}{2} m |\mathbf{v}|^2 f' \, \mathrm{d}\mathbf{v} \, \mathrm{d}\mathbf{x}$$

can take either sign. A Lyapunov functional must be nonnegative and vanish only at the equilibrium $f' \equiv 0$. The quadratic term

$$\iint \frac{f'^2}{W} \, \mathrm{d}\mathbf{v} \, \mathrm{d}\mathbf{x}$$

is positive definite when $W > 0$, thereby providing a consistent measure of the deviation from f_0.

A.2 Relation to Energy–Casimir (Relative Entropy) Structure

The form (.1) also arises naturally as the *second variation* of the free energy (or energy–Casimir functional)

$$\mathcal{F}[f, \mathbf{E}, \mathbf{B}] = \iint \Psi(f) \, \mathrm{d}\mathbf{v} \, \mathrm{d}\mathbf{x} + \int \left(\frac{\varepsilon_0}{2} |\mathbf{E}|^2 + \frac{1}{2\mu_0} |\mathbf{B}|^2 \right) \mathrm{d}\mathbf{x}, \tag{.3}$$

where Ψ is a convex function chosen so that

$$\Psi''(f_0) = \frac{1}{W}.$$

Expanding $\mathcal{F}$ to second order in $(f', \mathbf{E}', \mathbf{B}')$ yields

$$\delta^2 \mathcal{F}[f_0](f', \mathbf{E}', \mathbf{B}') = \iint \frac{f'^2}{2W} \, \mathrm{d}\mathbf{v} \, \mathrm{d}\mathbf{x} + \frac{\varepsilon_0}{2} \int |\mathbf{E}'|^2 \, \mathrm{d}\mathbf{x} + \frac{1}{2\mu_0} \int |\mathbf{B}'|^2 \, \mathrm{d}\mathbf{x}. \tag{.4}$$

Up to the constant factor $1/2$, this coincides with (.1).

A.3 Maxwellian Example

For a Maxwellian background

$$f_0(\mathbf{v}) = C \exp\bigl(-m|\mathbf{v}|^2\bigr)\,,$$

the natural choice for Ψ is the relative entropy

$$\Psi(f) = T\, f \log\left(\frac{f}{f_0}\right),$$

whose quadratic expansion yields $W = f_0/T$. In this case, $\iint f'^2/W$ corresponds to the Gaussian "distance" between f and f_0.